



May 28, 2026

From: NAVFAC MIDLANT PWD Earle FEAD

To: Regional Comptroller Office

Subj: **PROJECT SCOPE OF WORK**

Project Title: Inspection and Testing of Electric Fire pump and Motor/Engine, Diesel Fire Pumps, Controllers, and Oil/Water Separator at Building R38

Work Location: All work is located inside a room at building R38 in Naval Weapons Station Earle, 801 Hwy 36, Leonardo, New Jersey.

The room is a 31ftx41ft fire pump room located at Pier 3A in Naval Weapons Station Earle in Leonardo, NJ. The building must remain operational during the project. The contractor shall coordinate daily access to the building with the Contracting Officer prior to entry.

1. Objective and Scope

The contractor shall provide all labor, materials, equipment, and supervision to perform a complete inspection, operational test of the designated electric fire pump and motor, diesel fire pump, controllers and Oil/Water separator. All work must comply with the most current editions of all applicable regulations and standards. The objective is to ensure operational readiness, life safety compliance, and adherence to National Fire Protection Association (NFPA) and Unified Facilities Criteria (UFC) standards. All work shall conform to the safety requirements set forth in the most recent version of the Occupational Safety and Health Administration (OSHA), and Army Corps of Engineering EM 385-1-1 unless specified otherwise.

2. Reference Standards

All testing, inspection and disposal activities shall adhere to the following regulatory standards:

- NFPA 20: Standard for the Installation of Stationary Pumps for Fire Protection
- NFPA 25: Standard for the Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems
- UFC 3-601-02: Operations and Maintenance: Inspection, Testing, and Maintenance of Fire Protection Systems
- UFC 3-460-01: Design: Petroleum Fuel Facilities
- EPA Clean Water Act (CWA)
- Resource Conservation and Recovery Act (RCRA)
- Occupational Safety and Health Administration (OSHA) standards

- ## 3. Scope of Work: Diesel Fire pump, Electric Fire Pumps and Motor/Engines, controllers, and associated assemblies:
- The inspection and testing shall include, but is not limited to, the following: Due to the fire-damaged, the diesel equipment and the prolonged inactivity of the electric equipment, the contractor must complete all Pre-Start Diagnostics prior to energizing or operating the pumps. If critical faults are found during pre-start checks, the contractor shall notify the COR immediately and proceed to



the required inspection method based on discovered findings. The Inspection and Testing shall include but not limited to:

3.1. Visual and Fire Damage Assessment:

- 3.1.1. Inspect the diesel pump, the electric pump and motor/engine, controllers, and piping for physical damage, corrosion, or leaks.
- 3.1.2. Inspect all rubber components including but not limited to belts, hoses, seals, gaskets for heat degradation, melting, or brittleness.
- 3.1.3. Conduct a borescope inspection to ensure no soot, ash, or debris was ingested during the recent fire.
- 3.1.4. Verify all valves are in the correct open/closed position and secured.

3.2. Electrical & Controller Pre-Start Diagnostics:

- 3.2.1. Inspect the internal contacts of the electric fire pump controller for corrosion, pitting, or seizing due to prolonged inoperativeness.
- 3.2.2. Conduct an insulation resistance test on the electric motor windings and feeder cables prior to energizing the equipment to verify insulation integrity after prolonged inactivity.
- 3.2.3. Inspect the dual battery starting system, clean terminals, check electrolyte levels, and verify the operation and charging rate of the battery chargers.
- 3.2.4. Confirm the engine block heater is functioning.

3.3. Mechanical Pre-Start Inspection and Diagnostics:

- 3.3.1. Manually inspect both the electric and diesel pump shafts to ensure bearings are free-moving and impellers are not seized prior to applying power. Remove and inspect both diesel and electric pumps shaft as required. The shafts are at least 25ft deep from R38 floor level into the Bay. It is strongly recommended that the contractor should verify all equipment and dimensions in field prior to bidding the contract to avoid unforeseen site conditions.
- 3.3.2. Inspect shaft alignment, coupling, and guards.
- 3.3.3. Check lubrication of pump and motor bearings.
- 3.3.4. Verify the cooling water line to the engine is unobstructed.

3.4. Engine & Fuel System Inspection:

- 3.4.1. Inspect fuel lines, hoses, and filters. Check engine oil and coolant levels.
- 3.4.2. Inspect exhaust system for leaks or damage.
- 3.4.3. Conduct a visual inspection of the fuel quality and check for water in the fuel system. Verify the fuel tank is at least two-thirds (2/3) full.

3.5. No-Flow Operational Test:

- 3.5.1. Ensure a qualified technician is stationed directly at the controllers with an emergency stop plan during initial start-up, given the unknown internal condition of the equipment.
- 3.5.2. Run the pump under no-flow conditions for the minimum duration required by NFPA 25.

- 3.5.3. Perform baseline vibration analysis on the pump and motor/engine bearings during operation to identify internal wear, misalignment, or bearing degradation.
- 3.5.4. Utilize an infrared thermal camera during the test to monitor the electric motor casing, engine block, bearings, and electrical switchgear for abnormal hot spots.
- 3.5.5. Record starting pressure, churn pressure, motor voltage and amperage, engine RPM, oil
- 3.5.6. Pressure, and coolant temperature.
- 3.5.7. Verify packing gland performance and check for unusual noise or vibration.

3.6. Flow Operational Test:

- 3.6.1. Conduct a full flow test at 0%, 100%, and 150% of the pump's rated capacity using approved flow meters or hose streams.
- 3.6.2. Continue thermal imaging and vibration monitoring during peak load testing.
- 3.6.3. Plot the test results against the manufacturer's original pump curve. The fire pump shall be considered to have failed the test if it cannot deliver at least 95% of the flow and pressure showed on the original, unadjusted manufacturer's certified pump curve, as per NFPA 25.

3.7. Controller Operational Test:

- 3.7.1. Simulate a pressure drop to verify automatic start capabilities.
 - 3.7.2. Test manual start/stop functions.
 - 3.7.3. Verify the operation of all alarms and supervisory signals to the fire alarm control panel (FACP).
- 3.8. Upon completion of the inspection and testing, the contractor shall provide a comprehensive written report within ten (10) business days. The report shall include:
- 3.8.1. Completed Inspection Forms: Standardized NFPA 25 or DoD-equivalent inspection and testing forms.
 - 3.8.2. Performance Curve: A plotted graph comparing the current flow test results against the original factory pump curve.
- 3.9. Deficiency Report: A detailed list of any discrepancies, degraded components, or critical failures discovered during the inspection.
- 3.9.1. Fuel Quality Report: Results of any required diesel fuel testing, noting the presence of water, microbes, or degradation.
 - 3.9.2. Corrective Action Recommendations: Proposed solutions and cost estimates for required repairs to restore the system to full operational readiness.

4. Scope of Work: Inspection and Maintenance of Oil/Water Separator (OWS): The Inspection and Testing shall include but not limited to:

4.1. Phase 1: Comprehensive Inspection

- 4.1.1. The contractor shall perform a thorough inspection to assess the operational status and waste accumulation levels of the OWS.

- 4.1.2.** Secure the work area with appropriate barricades and signage.
- 4.1.3.** Conduct a visual site assessment of the area surrounding the OWS for any signs of leaks, spills, or structural issues.
- 4.1.4.** Implement Lockout/Tagout (LOTO) procedures for any associated electrical pumps or automated sensors before opening the OWS vault.

4.1.5. Internal Assessment:

- 4.1.5.1.** Measure and document the depth of the free-floating oil layer and the accumulated bottom sludge/sediment layer.
- 4.1.5.2.** Inspect the condition of the coalescing media (plates or tubes) for fouling or sediment buildup.
- 4.1.5.3.** Clean and test all electronic level sensors, float switches, and visual/audible alarms to ensure they accurately report high-oil or high-water levels.

4.1.6. Reporting and Decision:

- 4.1.6.1.** The contractor shall provide an immediate report of inspection findings to the Contracting Officer's Representative (COR).

4.1.7. The contractor shall execute Phase 2 if:

- 4.1.7.1.** The accumulated bottom sludge exceeds 50% of the separator's design capacity.
- 4.1.7.2.** The free-floating oil layer exceeds 80% of the separator's design capacity.
- 4.1.7.3.** The inspection reveals a critical malfunction or fouling that impedes proper operation.
- 4.1.7.4.** The service is required for the operation of the Oil-Water separator.

4.2. Phase 2: Cleaning and Servicing

4.2.1. Cleaning and Inspection:

- 4.2.1.1.** Perform high-pressure washing of the OWS interior walls, baffles, and coalescing media. Care must be taken not to damage the equipment.
- 4.2.1.2.** Extract all residual wash water and dislodged solids.
- 4.2.1.3.** Conduct a thorough visual inspection of the now-empty OWS. Check for cracks, structural degradation, corrosion, or leaks. Inspect internal baffles and piping for damage.

4.2.2. System Recharge:

- 4.2.2.1.** Upon completion of cleaning, refill the OWS with clean water to re-establish the proper operating level and water seal.
- 4.2.2.2.** Verify the system is ready to receive normal effluent.

- 4.3.** Upon completion of the service, the contractor must provide a comprehensive written report to the Contracting Officer Representative (COR) and the Environmental (EV) Office within ten (10) business days. The submission shall include:

- 4.3.1. Waste Manifests: Legible copies of all non-hazardous or hazardous waste manifests (e.g., EPA Form 8700-22) documenting the volume of waste removed, the transport entity, and the final authorized disposal or recycling facility.
- 4.3.2. Service Report: A detailed log of the cleaning process, including the date, time, volume of liquids/sludge removed, and the condition of the coalescing media.
- 4.3.3. Deficiency Report: A list of any structural defects, compromised baffles, or failing sensors identified during the inspection, accompanied by photographic evidence.
- 4.3.4. Corrective Action Recommendations: Proposed solutions and cost estimates for required repairs to restore the OWS to optimal condition.
- 4.3.5. The contractor must coordinate with the Contracting Office's Representative (COR) prior to the removal of any waste to ensure manifest sign-off procedures are followed correctly.
- 4.3.6. If personnel must enter the OWS vault for inspection, all work must strictly adhere to OSHA regulations, including continuous atmospheric monitoring and retrieval systems.
- 4.3.7. The contractor must maintain adequate spill response materials on-site during pumping/removal operation

5. Safety and Coordination

- 5.1 The Contractor shall coordinate with the Contracting Officer's Representative (COR) prior to taking the fire pump offline or initiating tests that may trigger facility alarms.
 - 5.2. Appropriate Lockout/Tagout (LOTO) procedures must be followed during electrical and mechanical inspections. Hearing protection shall be used during engine operation.
 - 5.3. Proper procedures must be followed for the handling and disposal of any hazardous materials, including diesel fuel, engine oil, coolant, and battery acid, in accordance with installation environmental policies.
- 6. All contaminated materials, including used absorbents, and recovered oil, coolant, and potentially contaminated fuel or cleaning materials shall be placed in clearly labeled, sealed containers. Contractor will be responsible for the transportation and disposal of all waste in accordance with all applicable regulations. The project is complete upon receipt of a successful clearance report and final approval from the Contracting Officer. The contractor shall include photo- as-found conditions in all reports.
 - 7. The government strongly recommends that prior to submitting a proposal the Contractor should make a site visit and become familiar with the field conditions, scope of the work, and all requirements for completing the contract scope. The Contractor should examine the site with respect to the work to be performed to ascertain all factors that would affect the cost of the work to the extent that such information is reasonably obtainable.
 - 8. The Contractor is responsible for initiating, maintaining, and supervising all safety precautions in connection with the Scope of Work. The Contractor shall coordinate with the Contracting Officer for utility supply disconnect. Requests shall be sent to the Contracting Officer three business days prior to the start of work.



9. Any damage to existing features to remain and new work shall be repaired or replaced in-kind at the Contractor's expense. All waste shall be disposed of in accordance with New Jersey Department of Environmental Protection's regulations.
10. Contractor's staging area is to be designated by Contracting Officer. Sidewalks area shall be kept clear of debris and open for use at all times. At the end of each workday, the Contractor shall clean the affected area so as not to pose a safety hazard.
11. The Contractor shall submit any/all submittals as required by the contracting officer for government approval, prior to the start of activities in this scope of work.
12. This project requires a Project Superintendent, a Quality Control (QCM) Manager and a Site Safety & Health Officer (SSHO). The same individual may be the Project Superintendent/Quality Control (QCM) Manager/ and Site Safety & Health Officer. However, that individual shall meet the qualification requirements of each position. The Contractor shall have at least one qualified supervisor capable of reading, writing, and conversing fluently in the English language on the job site during working hours. In addition, the Quality Control (QC) representative shall also have fluent English communication skills. SSHO shall meet the qualifications as outlined in EM 385-1-1 Appendix A, Section 1 and shall demonstrate a minimum of two years of experience as a project Safety Representative with no loss time incidents on assigned projects. Contractor shall provide the COR with a statement of qualifications in a letter to the Government for the SHHO.
13. Prior to the start of work, conference shall be attended by the Contractor's project manager/ the site superintendent. Coordinate the conference with the Contracting Officer's representative a minimum of five (5) business days in advance. The conference is required to discuss various aspects of the project and specific security and safety regulations at Naval Weapons Station Earle.
14. The Contractor shall submit labor hour's log for the Contracting officer's review and record. The Log can be submitted from start to end of the project.
15. The Contractor and his employees shall become acquainted with and obey all Government regulations as posted, or as requested by the Contracting Officer. Smoking is prohibited unless it is designated as smoking area.
16. The Government's regular (normal) working hours are from 0700 to 1530 (7:00 a.m. to 3:30 p.m.) Mondays through Fridays except on (a) federal holidays and (b) other days specifically designated by the Contracting Officer. Federal holidays are New Year's Day, Martin Luther King, Jr. Day, Presidents Day, Memorial Day, Juneteenth, Independence Day, Labor Day, Columbus Day, Veterans Day, Thanksgiving Day, and Christmas Day.
The Contractor will not be allowed to escort deliveries and workers into base. All Contractors entering the Base will need a badge. The Contractor shall obtain access to the installation by participating in the Defense Biometric Identification System (DBIDS), or by obtaining passes each day from the Base Pass and Identification Office. There is no cost for obtaining passes through the DBIDS. One-day passes, issued through the Base Pass and Identification Office, will be furnished without charge. The contractor shall submit all paperwork required by security for the employee access to the COR. The process of have a person cleared for access may take 14 business days. DBIDS Program: DBIDS is a voluntary program in which Contractor personnel who enroll, and are approved, are subsequently granted access to the



installation for a period up to one year, or the length of the contract, whichever is less, and are not required to obtain a new pass from the Base Pass and Identification Office for each visit. The Government performs background screening and credentialing. Throughout the year the Contractor employee must continue to meet background screening standards. Periodic background screenings are conducted to verify continued DBIDS participation and installation access privileges. Under the DBIDS program, commercial vehicle inspection is required, including situations of Random Anti-Terrorism Measures (RAM) or in the case of an elevation of Force Protection Conditions (FPCON). Information on requirements to participate and enroll in DBIDS is available at <http://dbids.dmdc.mil/> or by calling 1-800-372-7437. One-Day Passes: Participation in the DBIDS is not mandatory, and if the Contractor chooses to not participate, the Contractor's personnel will have to obtain daily passes, be subject to daily mandatory vehicle inspection, and will have limited access to the installation. The Government will not be responsible for any cost or lost time associated with obtaining daily passes or added vehicle inspections incurred by non-participants in the DBIDS. Work outside regular working hours requires Contracting Officer approval. Project shall be completed 8 weeks from the date of award.

17. Attachment:

17.1. Pictures

17.2. Existing drawings: Supplementary for reference only

Azeb Hailu, P.E.
Engineering Tech.
Public Works Department Earle
201 State Highway 34, Bldg. C-9, Colts Neck, NJ 07722
Office: 732-866-2396, Cell: 617-943-7796
Email azeb.a.hailu.civ@us.navy.mil

17.1. Pictures:

Picture-1: Deisel Pump system after fire damage



Picture-2: Electric pump system





17.2. Existing Drawings: All existing drawings are supplementary for reference only

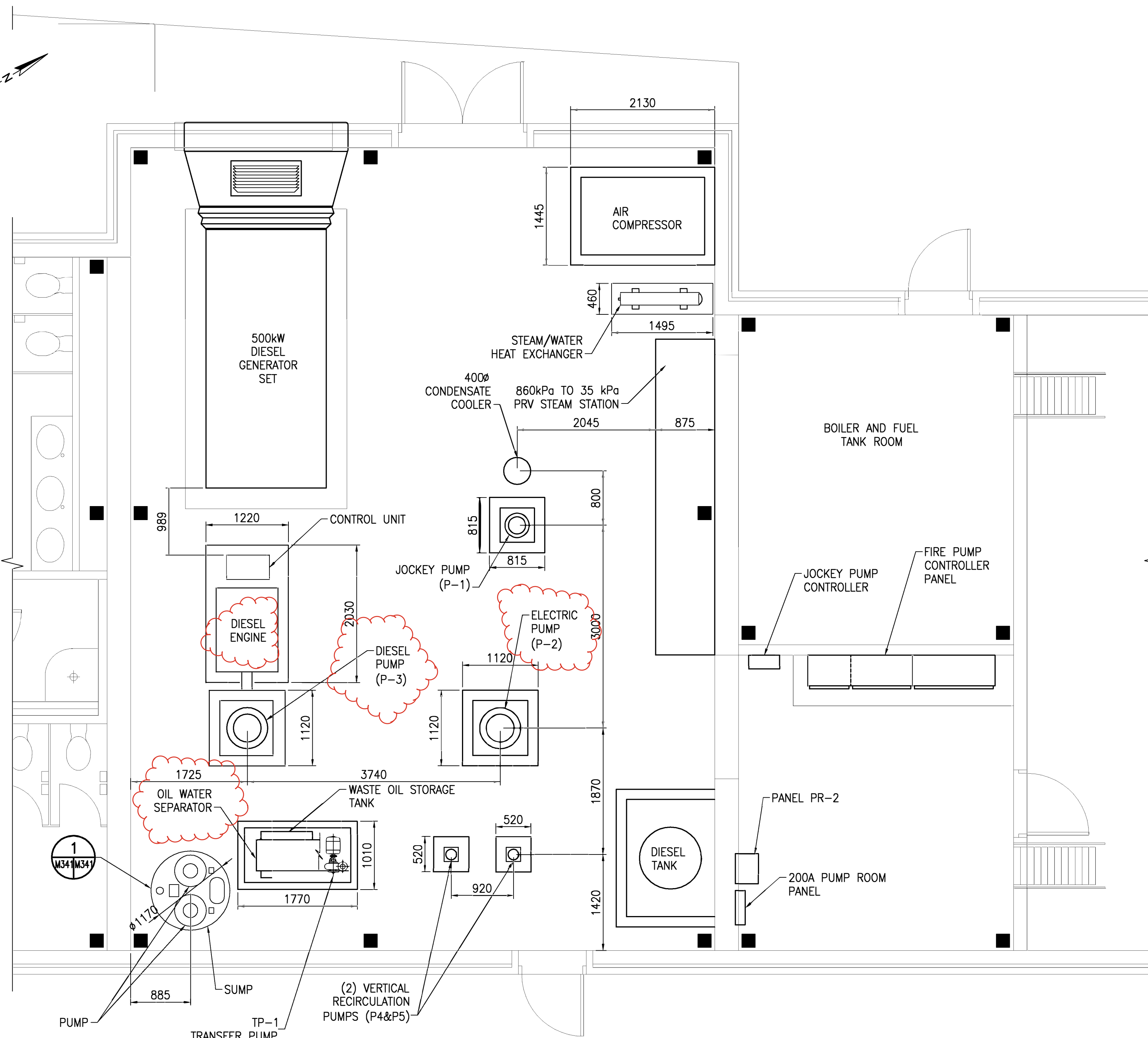
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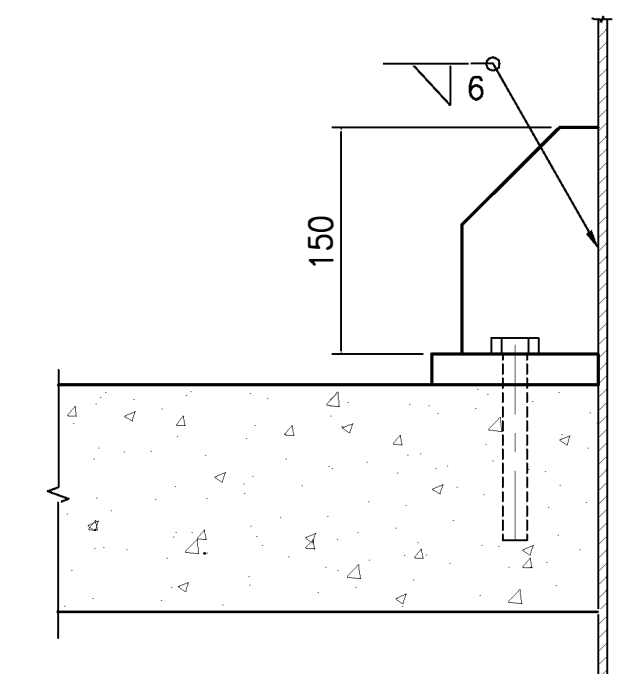
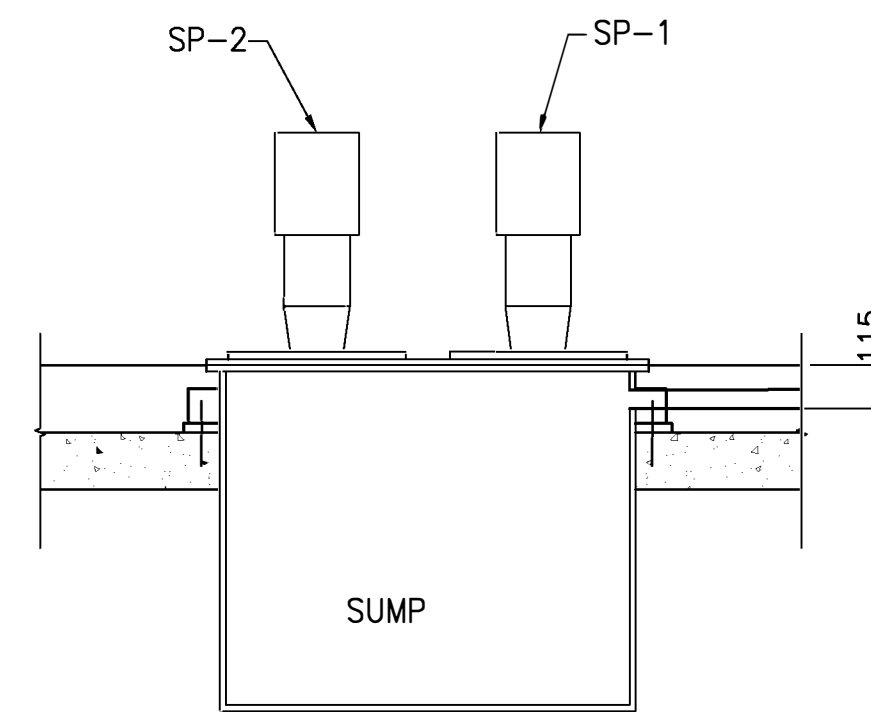
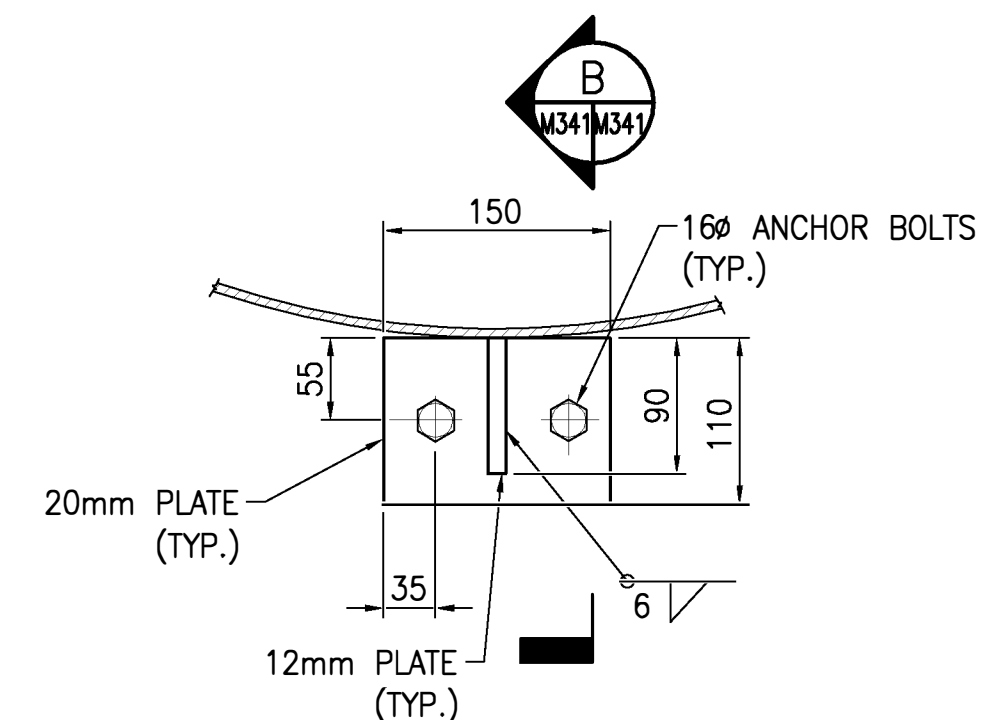
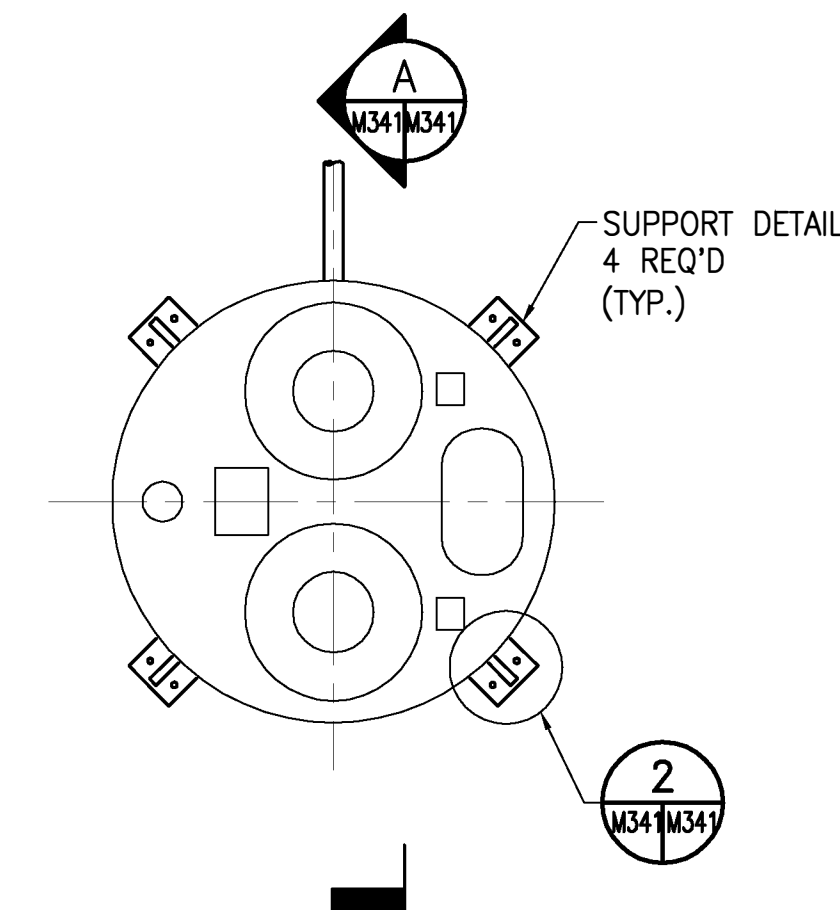
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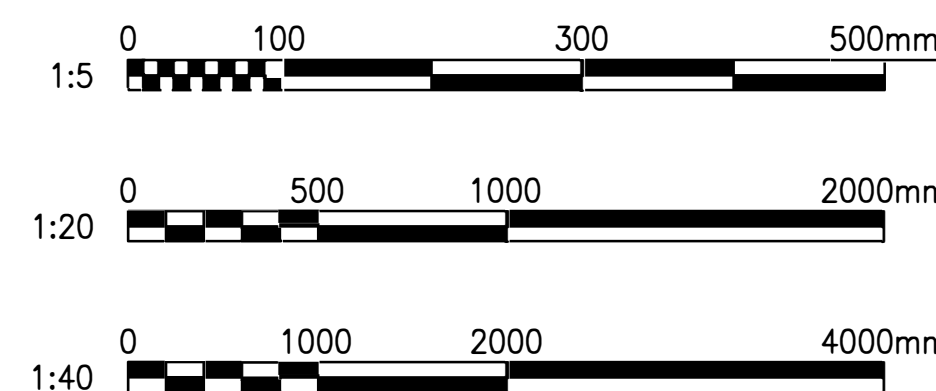


FIRE PUMP ROOM PLAN
1:40



NOTE

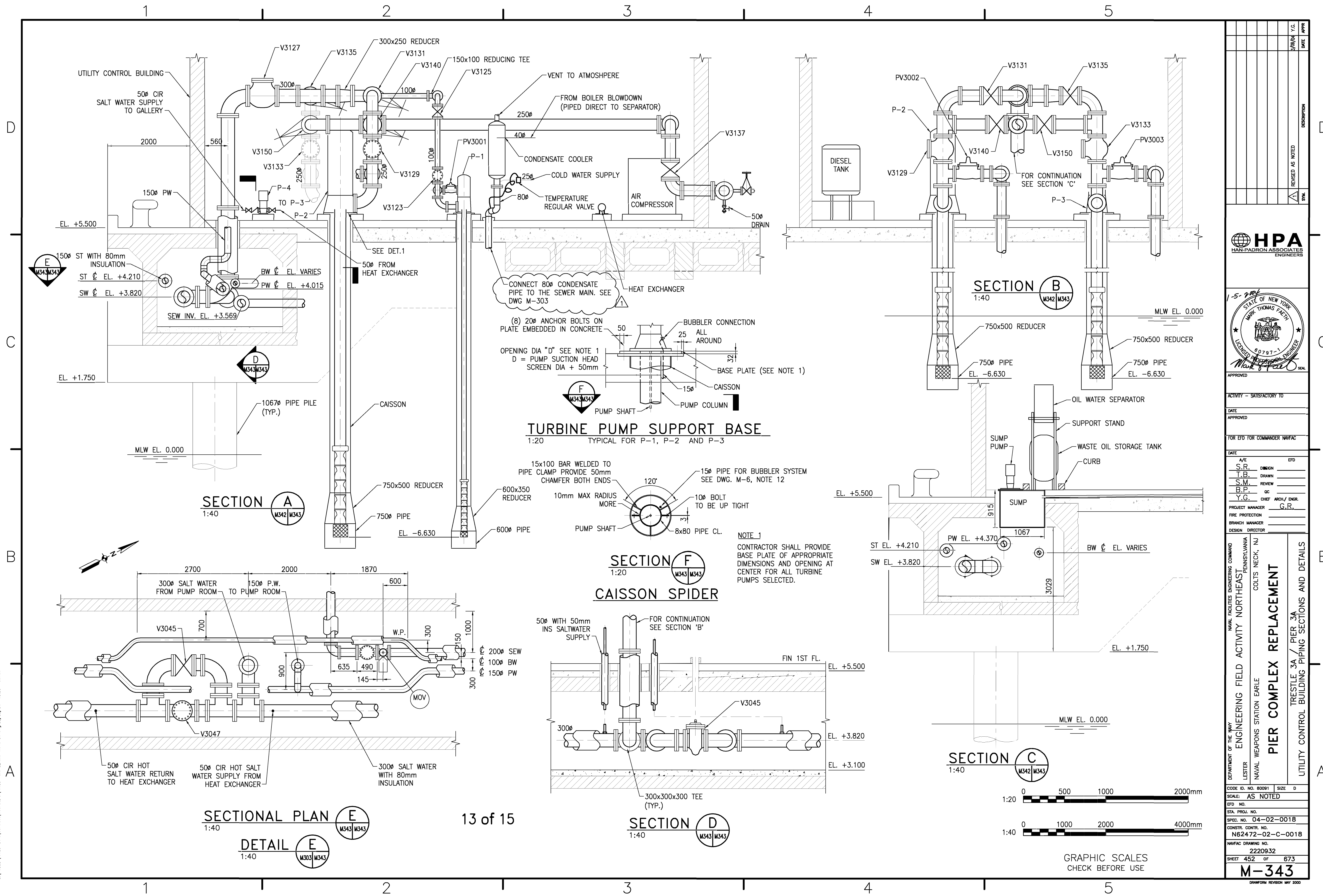
1. FOR PUMP CONTROLLERS SEE DWG. NO. E-344.
2. FOR EQUIPMENT SCHEDULE SEE DWG. NO. M-347.
3. CONTRACTOR TO VERIFY ALL DIMENSIONS PRIOR TO CONSTRUCTION.
4. FINAL CONNECTIONS TO SELECTED EQUIPMENT IS RESPONSIBILITY OF CONTRACTOR.



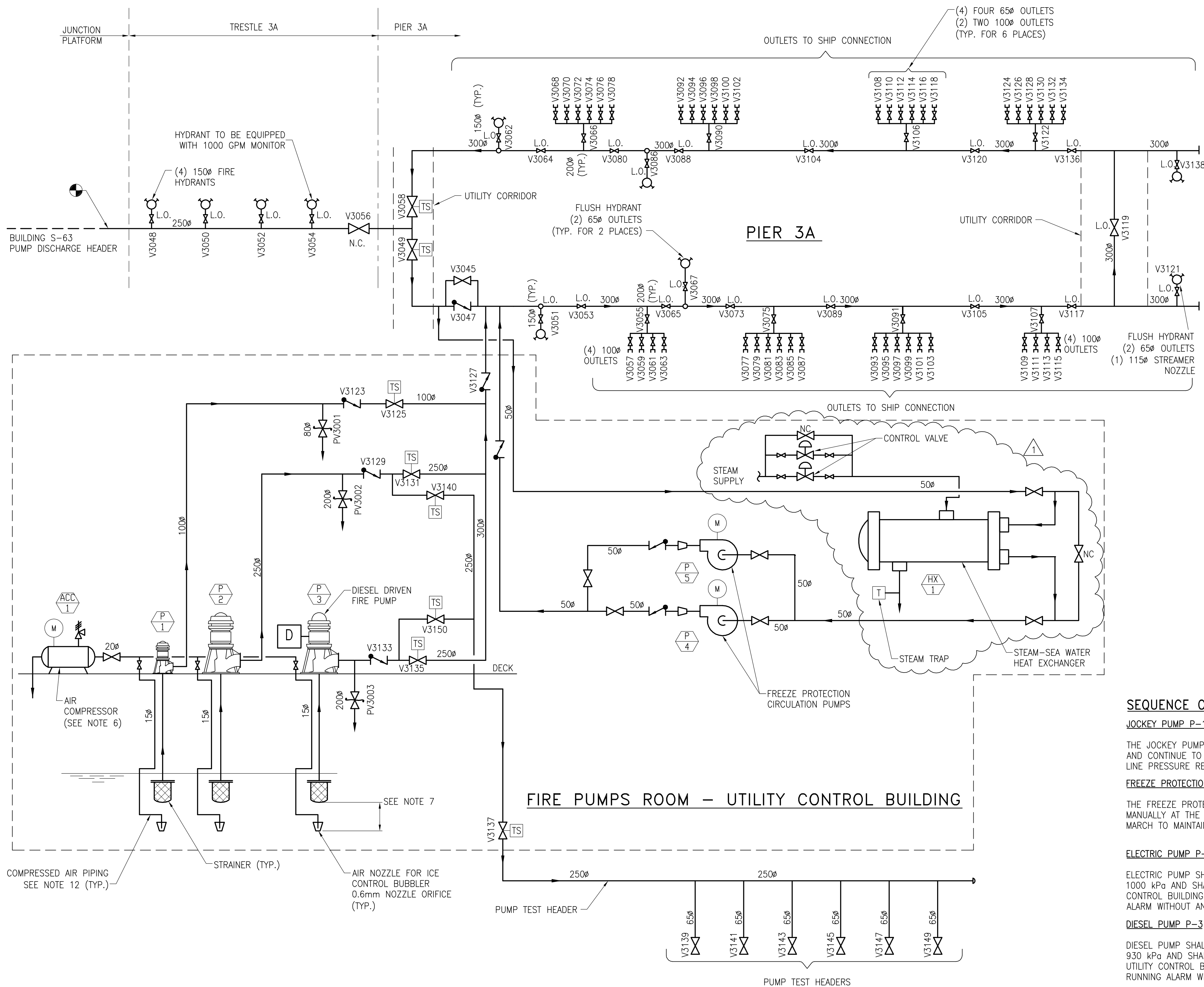
GRAPHIC SCALES
CHECK BEFORE USE

DATE		APPR
SYN		DESCRIPTION
HPA HAN-PADRON ASSOCIATES ENGINEERS		
1-5-2004		
Mark S. Fazio APPROVED		
ACTIVITY - SATISFACTORY TO		
DATE		
APPROVED		
FOR EFD FOR COMMANDER NAVFAC		
DATE	A/E	EFD
	A.M.	DESIGN
	T.B.	DRAWN
	S.M.	REVIEW
	B.P.	QC
	Y.G.	CHIEF ARCH/ENGR.
PROJECT MANAGER		
FIRE PROTECTION		
BRANCH MANAGER		
DESIGN DIRECTOR		
NAVAL FACILITIES ENGINEERING COMMAND ENGINEERING FIELD ACTIVITY NORTHEAST PENNSYLVANIA NAVAL WEAPONS STATION EARLE COLTS NECK, NJ		
PIER COMPLEX REPLACEMENT		
TRESTLE 3A / PIER 3A UTILITY CONTROL BUILDING EQUIPMENT ARRANGEMENT		
CODE ID. NO.	80091	SIZE D
SCALE: AS NOTED		
EFD NO.		
STA. PROJ. NO.		
SPEC. NO. 04-02-0018		
CONSTR. CONTR. NO.		
N62472-02-C-0018		
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NOTES

1. SALT/FIRE WATER MAIN AND BRANCH PIPES SHALL BE MADE OF DUCTILE IRON CLASS 50, CEMENT LINED, WITH MECHANICAL JOINTS, UNLESS OTHERWISE NOTED.
2. ALL PIPING, VALVES AND FITTINGS, SHALL BE MIN. 250# CLASS AND FREEZE PROTECTED.
3. MAIN AND BRANCH PIPING SHALL BE FREEZE PROTECTED BY USING A COMBINATION OF HEATED SALT WATER, PIPE FIBERGLASS INSULATION AND WITH ALL WEATHERPROOF JACKET.
4. OUTLET VALVES AND PIPING SHALL BE FREEZE PROTECTED BY USING A COMBINATION OF ELECTRIC HEAT TRACING, PIPE FIBER GLASS INSULATION AND WITH ALL WEATHER PROOF JACKET.
5. ALL PIPING LOCATED INSIDE UTILITY GALLERIES AND ACCESS CORRIDORS, SHALL BE SUPPORTED WITH PIPE ROLL HANGERS TYPE UNLESS OTHERWISE NOTED.
6. AIR COMPRESSOR UNIT 0.005 m³/s @ 860 kPa, 455 L TANK CAPACITY 4 kW.
7. AIR NOZZLES SHALL BE LOCATED 3 METERS DIRECTLY BELOW BOTTOM OF STRAINERS. NOZZLES SHALL DISCHARGE DOWN. PIPING SHALL BE PROPERLY SECURED TO WITHSTAND SEA CURRENTS, ICE AND DEBRIS.
8. FOR ELECTRICAL CONNECTIONS REFER TO DWG. E-344.
9. PUMP ROOM PIPING SHALL HAVE FLANGED CONNECTIONS.
10. CONTRACTOR SHALL BE RESPONSIBLE FOR PUMPS START/STOP PRESSURE SETTINGS, SO THAT, SALT WATER LOOP PRESSURE SHALL NOT EXCEED ITS RATING.
11. FOR EQUIPMENT SCHEDULE SEE DWG M-347.
12. ALL PIPING FOR THE BUBBLER SYSTEM SHALL BE MADE OF STAINLESS STEEL, GRADE A312. VERTICAL PIPING SHALL BE LOCATED INSIDE OF THE CAISSON AND PROPERLY FASTENED TO THE PIPE CLAMPS.

SEQUENCE OF OPERATION

JOCKEY PUMP P-1

THE JOCKEY PUMP SHALL START AUTO WHEN THE LINE PRESSURE DROPS TO 1034 kPa AND CONTINUE TO MAINTAIN THE LINE PRESSURE AT 1034 kPa. IT WILL STOP WHEN THE LINE PRESSURE REACHES 1103 kPa.

FREEZE PROTECTION CIRCULATION PUMPS P-4 & P-5

THE FREEZE PROTECTION CIRCULATION PUMPS WORKING ALTERNATELY SHALL BE STARTED MANUALLY AT THE BEGINNING OF NOVEMBER AND SHALL BE STOPPED AT THE END OF MARCH TO MAINTAIN THE FIREMAIN WATER TEMPERATURE ABOVE FREEZING.

ELECTRIC PUMP P-2

ELECTRIC PUMP SHALL START AUTO WHEN THE FIREMAIN LINE PRESSURE DROPS TO 1000 kPa AND SHALL BE STOPPED MANUALLY AT PUMP P-2 CONTROLLER IN UTILITY CONTROL BUILDING BY FIREHOUSE ATTENDANT IN RESPONSE TO PUMP P-2 RUNNING ALARM WITHOUT ANY FIRE.

DIESEL PUMP P-3

DIESEL PUMP SHALL START AUTO WHEN THE FIREMAIN LINE PRESSURE DROPS TO 930 kPa AND SHALL BE STOPPED MANUALLY AT PUMP P-3 CONTROLLER IN UTILITY CONTROL BUILDING BY FIREHOUSE ATTENDANT IN RESPONSE TO PUMP P-3 RUNNING ALARM WITHOUT ANY FIRE.

TRESTLE 3A/PIER 3A - SALT/FIRE WATER SYSTEM FLOW DIAGRAM
N.T.S.

REV	DATE	BY	APP	DESCRIPTION
1	02/23/04	Y.G.		REVISED STEAM SUPPLY
2				
3				
4				
5				

APPROVED	ACTIVITY - SATISFACTORY TO
DATE	APPROVED
FOR EFD FOR COMMANDER NAVFAC	
DATE	
A/E	EFD
J.U.	DESIGN
T.B.	DRAWN
S.M.	REVIEW
B.P.	QC
Y.G.	CHIEF ARCH/ ENGR.
PROJECT MANAGER	
FIRE PROTECTION	
BRANCH MANAGER	
DESIGN DIRECTOR	

NAVAL FACILITIES ENGINEERING COMMAND
ENGINEERING FIELD ACTIVITY NORTHEAST
PENNSYLVANIA
COLTS NECK, NJ
NAVAL WEAPONS STATION EARLE
PIER COMPLEX REPLACEMENT
FLOW DIAGRAM
SALT/FIRE WATER SYSTEM

CODE ID. NO. 80091	SIZE D
SCALE: NONE	
EFD NO.	
STA. PROJ. NO.	
SPEC. NO. 04-02-0018	
CONSTR. CONTR. NO. N62472-02-C-0018	
NAVFAC DRAWING NO. 2220890	
SHEET 410 OF 673	
M-6	
DRAWING REVISION MAY 2000	

PUMP SCHEDULE										
PUMP NO.	SERVICE	FLOW L/min	DISCHARGE kPa	MOTOR			RPM	PUMP TYPE	LOCATION	REMARKS
				kW	VOLT	ø/HZ				
PSEW-1	SEWER	2080	TDH 380	37	460	3/60	1755	CENTRIFUGAL	M-330	LIFT STATION PIER 3A
PSEW-2	SEWER	2080	TDH 380	37	460	3/60	1755	CENTRIFUGAL	M-330	LIFT STATION PIER 3A
PSMP-1	SEWER	70	TDH 60	0.25	115	1/60	1700	CENTRIFUGAL	M-330	LIFT STATION PIER 3A
SP-1/2	OIL/WATER	95	70	0.37	460	3/60	1700	CENTRIFUGAL	M-340	SUMP PUMP
TP-1	WASTE OIL	75	210	0.75	460	3/60	1700	CENTRIFUGAL	M-340	TRANSFER PUMP
PBW-1	BILGE WATER	1370	95	4	460	3/60	<900	POS. DISPL.	M-403	LIFT STATION (BID OPTION 5)
PBW-2	BILGE WATER	1370	95	4	460	3/60	<900	POS. DISPL.	M-403	LIFT STATION (BID OPTION 5)
PBW-3	BILGE WATER	1370	260	10	460	3/60	<900	POS. DISPL.	M-414	LIFT STATION (BID OPTION 8)
PBW-4	BILGE WATER	1370	260	10	460	3/60	<900	POS. DISPL.	M-414	LIFT STATION (BID OPTION 8)
PBW-5	BILGE WATER	1370	95	4	460	3/60	<900	POS. DISPL.	M-415	LIFT STA. ONSHORE (BID OP. 8)
PBW-6	BILGE WATER	1370	95	4	460	3/60	<900	POS. DISPL.	M-415	LIFT STA. ONSHORE (BID OP. 8)
BP-1	BILGE WATER	155	60	0.3	460	3/60	<900	POS. DISPL.	M-406	BOOSTER PUMP PIER 4
BP-2	BILGE WATER	155	60	0.3	460	3/60	<900	POS. DISPL.	M-406	BOOSTER PUMP PIER 3A

FIRE PUMP SCHEDULE										
PUMP NO.	SERVICE	FLOW L/min	DISCHARGE kPa	MOTOR			RPM	PUMP TYPE	LOCATION	REMARKS
				kW	VOLT	ø /HZ				
P-1	S/F WATER	230	1103	15	460	3/60	3600	VERTICAL	M-340	JOCKEY PUMP
P-2	S/F WATER	5680	1140	185	460	3/60	1800	VERTICAL	M-340	FIRE PUMP/FC
P-3	S/F WATER	5680	1140	DIESEL DRIVEN	-	-	-	VERTICAL	M-340	FIRE PUMP/FC
P-4	S/F WATER	115	70	0.75	460	3/60	1800	CENTRIFUGAL	M-340	CIRC. PUMP
P-5	S/F WATER	115	70	0.75	460	3/60	1800	CENTRIFUGAL	M-340	CIRC. PUMP

WASTE OIL/WATER SUMP WITH PUMPS AND CONTROLS				
INSIDE DIA OF SUMP (m)	OVERALL DIA OF COVER (m)	DUPLEX PUMPS	STEEL THICKNESS	HEIGHT (m)
1.06	1.17	SP-1/2	16	0.915

OWS (OIL/WATER SEPARATOR)							
FLOW (L/min)	HOLDING CAPACITY (L)	LENGTH (m)	WIDTH (m)	HEIGHT (m)	INLET & OUTLET (mm)	OIL DRAW OFF (mm)	VENT (mm)
245	378	1.02	0.76	0.91	100	50	100

OIL STORAGE TANK				
CAPACITY (L)	LENGTH (m)	WIDTH (m)	HEIGHT (m)	VENT (mm)
520	0.76	0.69	1.13	50

DIESEL TANK (DOUBLE WALL)				
CAPACITY (L)	HEIGHT (m)	DIA (m)	MAT'L (m)	THICKNESS (mm)
910	1.22	0.97	CS	2.77

FAN SCHEDULE									
FAN NO.	SERVICE	FLOW M3/HR	TOTAL STATIC (Hz0 mm)	MOTOR			RPM	FAN TYPE	REF. DWG
				kW	VOLT	ø/HZ			
EF-1	LIFT STATION EXHAUST FAN	1390	16	0.2	115	1/60	1755	CENTRIFUGAL DOWN BLAST	M-330
EF-2	DRUM SCRUBBER FAN	199	51	0.37	115	1/60	3450	CENTRIFUGAL	M-330

CONDENSATE COOLER						
MARK	LOCATION	QTY	MFGR.	OPERATING PRESSURE	OPERATING TEMP.	REMARKS
	UTILITY CONTROL BLDG. PIER 3A	1	APPROVED	1035 kPa	200 °C	---

DUPLIX AIR COMPRESSOR								
MARK	LOCATION	MODEL #	DRIVER WATTS EACH	RECEIVER CAPACITY (L)	CMM @ 1208 kPa	RPM	ELECTRICAL	REMARKS
AC-1	UTILITY CONTROL BLDG. PIER 3A	APPROVED MNFG.	1490	227	0.2	510	208V/3Ø/60HZ	--

DIESEL ENGINE						
COMBUSTION AIR @ 15.5°C (m.3/s)	HEAT REJECTION TO AIR (kW)	MAX FUEL CONSUMPTION (L/min)	LENGTH (m)	WIDTH (m)	POWER (kW)	REMARKS
0.27	30	0.88	1.57	0.9	197.6	---

HEAT EXCHANGER FOR STEAM/SALT WATER						
FLOW (L/min)	TUBE ENTERING WATER TEMP. (°C)	TUBE LEAVING WATER TEMP. (°C)	MASS FLOW RATE FOR STEAM (kg/s)	DIA (m)	LENGTH (m)	REMARKS
115	4.44	10	0.02	0.184	1.01	---

[illegible]

1-5-2004

Mark S. Felt SEAL

ACTIVITY - SATISFACTORY TO

DATE _____

APPROVED

FOR EFD FOR COMMANDER NAVFAC

DATE _____

A/E		EFD
J.J.	DESIGN	
T.B.	DRAWN	
S.M.	REVIEW	
B.P.	QC	
Y.G.	CHIEF ARCH./ ENGR	

PROJECT MANAGER G.R.
FIRE PROTECTION _____
BRANCH MANAGER _____
DESIGN DIRECTOR _____

DEPARTMENT OF THE NAVY ENGINEERING FIELD ACTIVITY LESTER	NAVAL FACILITIES ENGINEERING COMMAND NORTH EAST PENNSYLVANIA	COLTS NECK, NJ
NAVAL WEAPONS STATION EARLE		
PIER COMPLEX REPLACEMENT		
TRISTLE 3A / PIER 3A EQUIPMENT SCHEDULE		

CODE ID. NO. 80091		SIZE	D
SCALE: NONE			
EFD NO.			
STA. PROJ. NO.			
SPEC. NO. 04-02-0018			
CONSTR. CONTR. NO. N62472-02-C-0018			
NAVFAC DRAWING NO. 2220936			
SHEET 456		OF	673
M-347			
DRAWFORM REVISION MAY 2000			